

Chapter 6

More on Matrices

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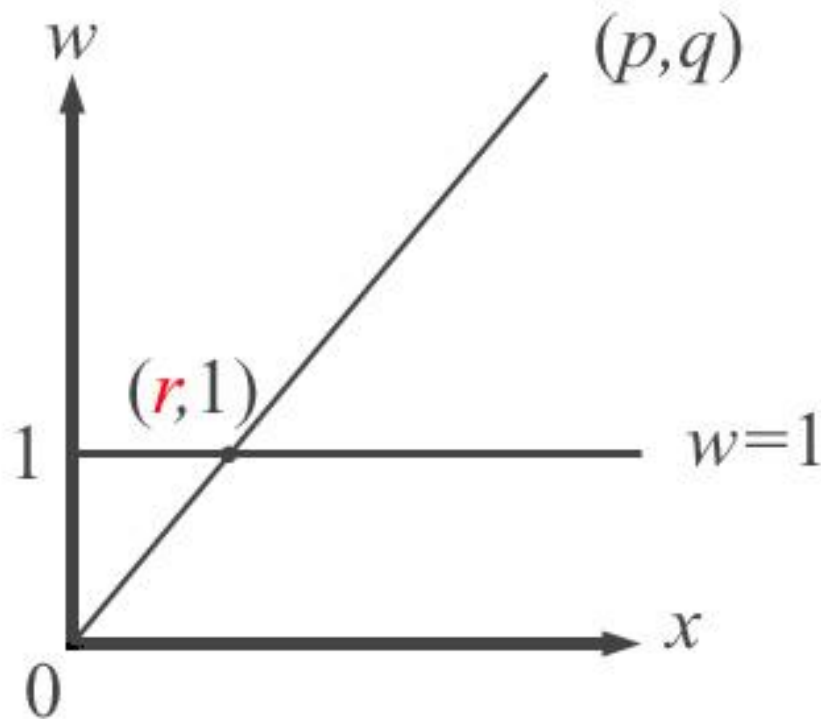
Homogenous Coordinates

- Extend 3D into 4D.
- The 4th dimension is not “time”.
- The 4th dimension is really just a kluge to help the math work out.
- The 4th dimension is called w .

Extending 1D into Homogenous Space

- Start with 1D, its easier to visualize than 3D.
- Homogenous 1D coords are of the form (x, w) .
- Imagine the vanilla 1D line lying at $w = 1$.
- So the 1D point x has homogenous coords $(x, 1)$.
- Given a homogenous point (x, w) , the corresponding 1D point is its projection onto the line $w = 1$ along a line to the origin, which turns out to be $(x/w, 1)$.

What are the 2D Coords of Homogenous Point (p,q) ?



Projecting Onto 1D Space

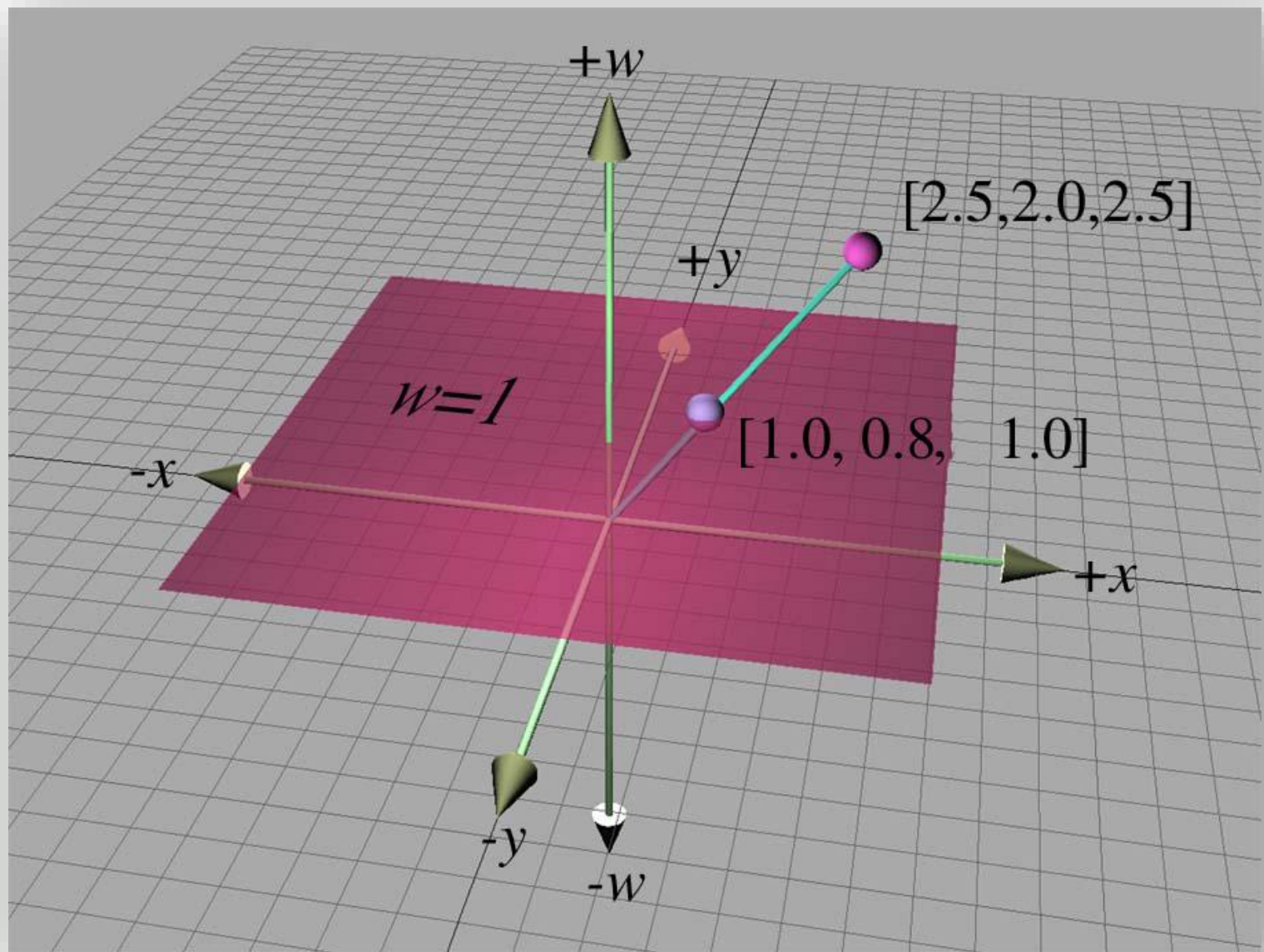
- Each point x in 1D space corresponds to an infinite number of points in homogenous space, those on the line from the origin through the point $(x, 1)$.
- The homogenous points on this line project onto its intersection with the line $w = 1$.

Extending 2D into Homogenous Space

- 2D next, it's *still* easier to visualize than 3D.
- Homogenous 2D coordinates are of the form (x, y, w) .
- Imagine the vanilla 2D plane lying at $w = 1$.
- So the 2D point (x, y) has homogenous coordinates $(x, y, 1)$.

Projecting Onto 2D Space

- Each point (x, y) in 2D space corresponds to an infinite number of points in homogenous space.
- Those on the line from the origin thru $(x, y, 1)$.
- The homogenous points on this line project onto its intersection with the plane $w = 1$.



2D Homogenous Coordinates

- Just like before (argument omitted), the homogenous point (x, y, w) corresponds to the 2D point $(x/w, y/w, 1)$.
- That is, the 2D equivalent of the homogenous point (p, q, r) is $(p/r, q/r)$.

3D Homogenous Coordinates

- This extends to 3D in the obvious way.
- The homogenous point (x, y, z, w) corresponds to the 3D point $(x/w, y/w, z/w, 1)$.
- That is, the 3D equivalent of the homogenous point (p, q, r, s) is $(p/s, q/s, r/s)$.

Point at Infinity

- w can be any value except 0 (divide by zero error).
- The point $(x,y,z,0)$ can be viewed as a “point at infinity”

Why Use Homogenous Space?

- It will let us handle translation with a matrix transformation.
- Embed 3D space into homogenous space by basically ignoring the w component.
- Vector (x, y, z) gets replaced by $(x, y, z, 1)$.
- Does that “1” at the end sound familiar?

Homogenous Matrices

Embed 3D transformation matrix into 4D matrix by using the identity in the w row and column.

$$\begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix} \Rightarrow \begin{bmatrix} m_{11} & m_{12} & m_{13} & 0 \\ m_{21} & m_{22} & m_{23} & 0 \\ m_{31} & m_{32} & m_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4D Matrix Multiplication

$$\begin{bmatrix} x & y & z & 1 \end{bmatrix} \begin{bmatrix} m_{11} & m_{12} & m_{13} & 0 \\ m_{21} & m_{22} & m_{23} & 0 \\ m_{31} & m_{32} & m_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} xm_{11} + ym_{21} + zm_{31} \\ xm_{12} + ym_{22} + zm_{32} \\ xm_{13} + ym_{23} + zm_{33} \\ 1 \end{bmatrix}^T$$

Translation Matrices

Kluge 3D translation matrix by shearing 4D space.

$$\begin{bmatrix} x & y & z & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \Delta x & \Delta y & \Delta z & 1 \end{bmatrix}$$
$$= \begin{bmatrix} x + \Delta x & y + \Delta y & z + \Delta z & 1 \end{bmatrix}$$

Translation vs. Orientation

- Just like in 3D, compose 4D operations by multiplying the corresponding matrices.
- The translation and orientation parts of a composite matrix are independent.
- For example, let **R** be a rotation matrix and **T** be a translation matrix.
- What does **M = RT** look like?

$$\mathbf{R} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{T} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \Delta x & \Delta y & \Delta z & 1 \end{bmatrix}$$

Rotate then Translate

- Then we could rotate and then translate a point $\mathbf{v}$ to a new point $\mathbf{v}'$ using $\mathbf{v}' = \mathbf{vRT}$.
- We are rotating first and then translating.
- The order of transformations is important, and since we use row vectors, the order of transformations coincides with the order that the matrices are multiplied, from left to right.

$M = RT$

Just as with 3 x 3 matrices, we can concatenate the two matrices into a single transformation matrix, which we'll call **M**.

Let **M** = **RT**, so

$$\mathbf{v}' = \mathbf{v}RT = \mathbf{v}(RT) = \mathbf{vM}$$

$$\mathbf{M} = \mathbf{RT}$$

$$= \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ \Delta x & \Delta y & \Delta z & 1 \end{bmatrix}$$

$$= \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ \Delta x & \Delta y & \Delta z & 1 \end{bmatrix}$$

In Reverse

- Applying this information in reverse, we can take a 4 x 4 matrix **M** and separate it into a linear transformation portion, and a translation portion.
- We can express this succinctly by letting the translation vector **t** = $[\Delta x, \Delta y, \Delta z]$.

$$\mathbf{M} = \begin{bmatrix} \mathbf{R} & \mathbf{0} \\ \mathbf{t} & 1 \end{bmatrix}$$

Points at Infinity Again

- Points at infinity are actually useful.
- They orient just like points with $w \neq 0$: multiply by the orientation matrix.
- But they don't translate: translation matrices have no effect on them.

Matrix Without Translation

$$\begin{bmatrix} x & y & z & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$
$$= \begin{bmatrix} xr_{11} + yr_{21} + zr_{31} \\ xr_{12} + yr_{22} + zr_{32} \\ xr_{13} + yr_{23} + zr_{33} \\ 0 \end{bmatrix}^T$$

Matrix With Translation

$$\begin{bmatrix} x & y & z & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ \Delta x & \Delta y & \Delta z & 1 \end{bmatrix}$$
$$= \begin{bmatrix} xr_{11} + yr_{21} + zr_{31} \\ xr_{12} + yr_{22} + zr_{32} \\ xr_{13} + yr_{23} + zr_{33} \\ 0 \end{bmatrix}^T$$

} Same

So What?

- The translation part of 4D homogenous transformation matrices has no effect on points at infinity.
- Use points at infinity for things that don't need translating (eg. Surface normals).
- Use regular points (with $w = 1$) for things that *do* need translating (eg. Points that make up game objects).

4x3 Matrices

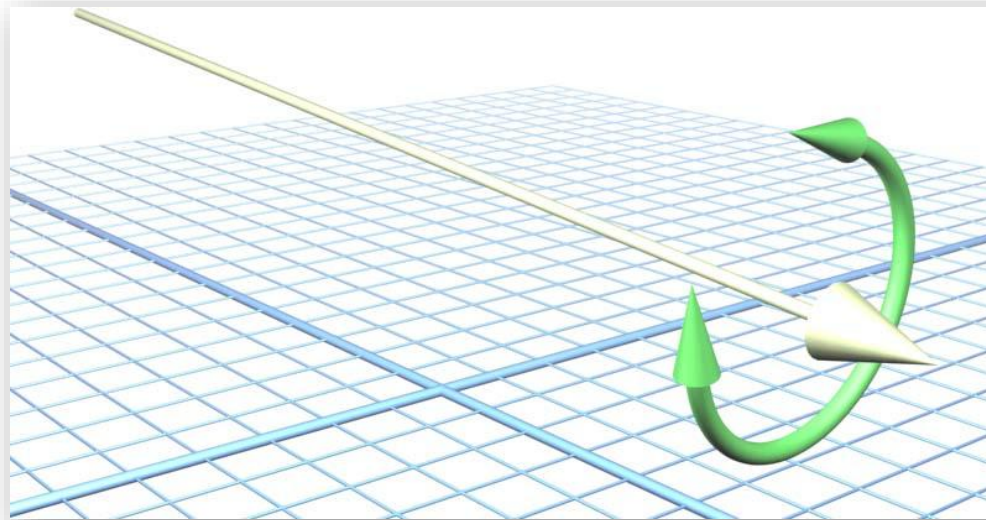
- The last column of 4D homogenous transformation matrices is always $[0, 0, 0, 1]^T$.
- Technically it always needs to be there for the algebra to work out.
- But we know what it's going to do, so there's no reason to implement it in code.

Section 8.1:

What Exactly is “Orientation”?

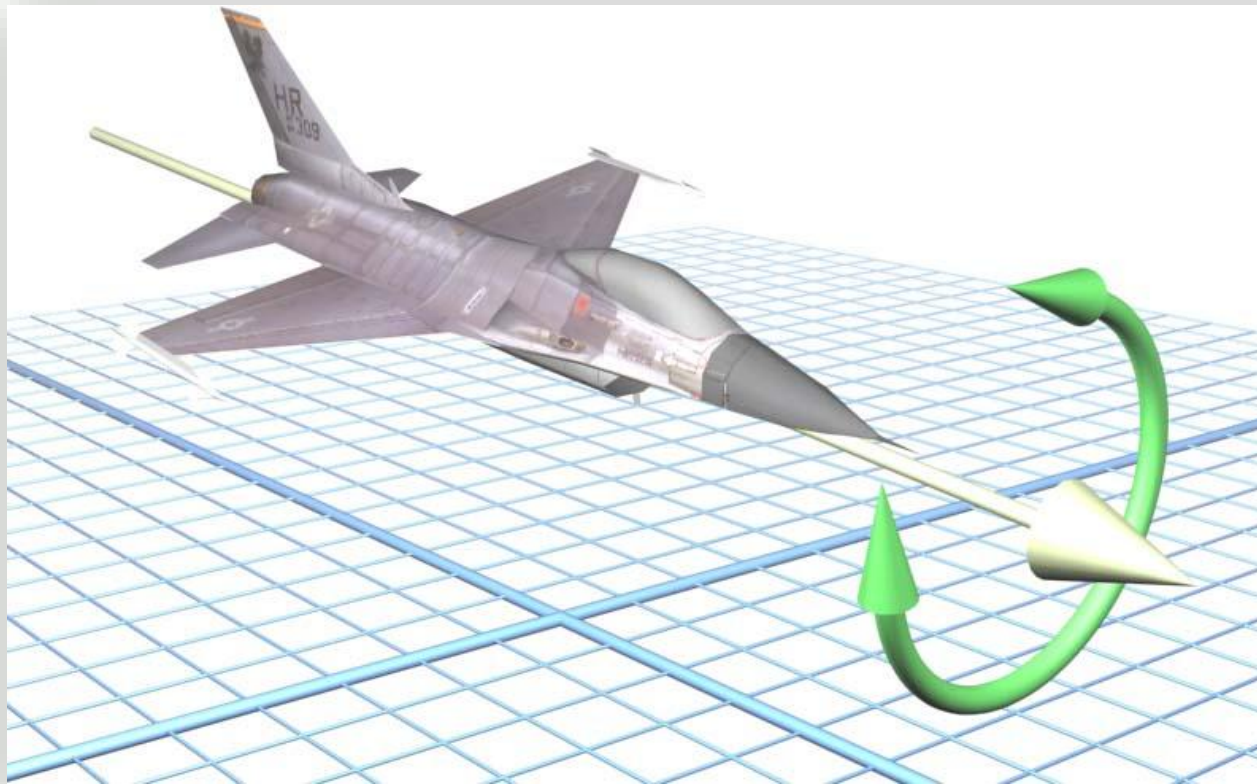
Orientation

- What is orientation? More than direction.
- A vector specifies direction but it can also be twisted.

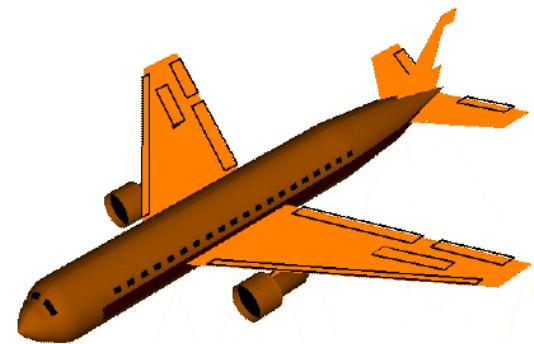
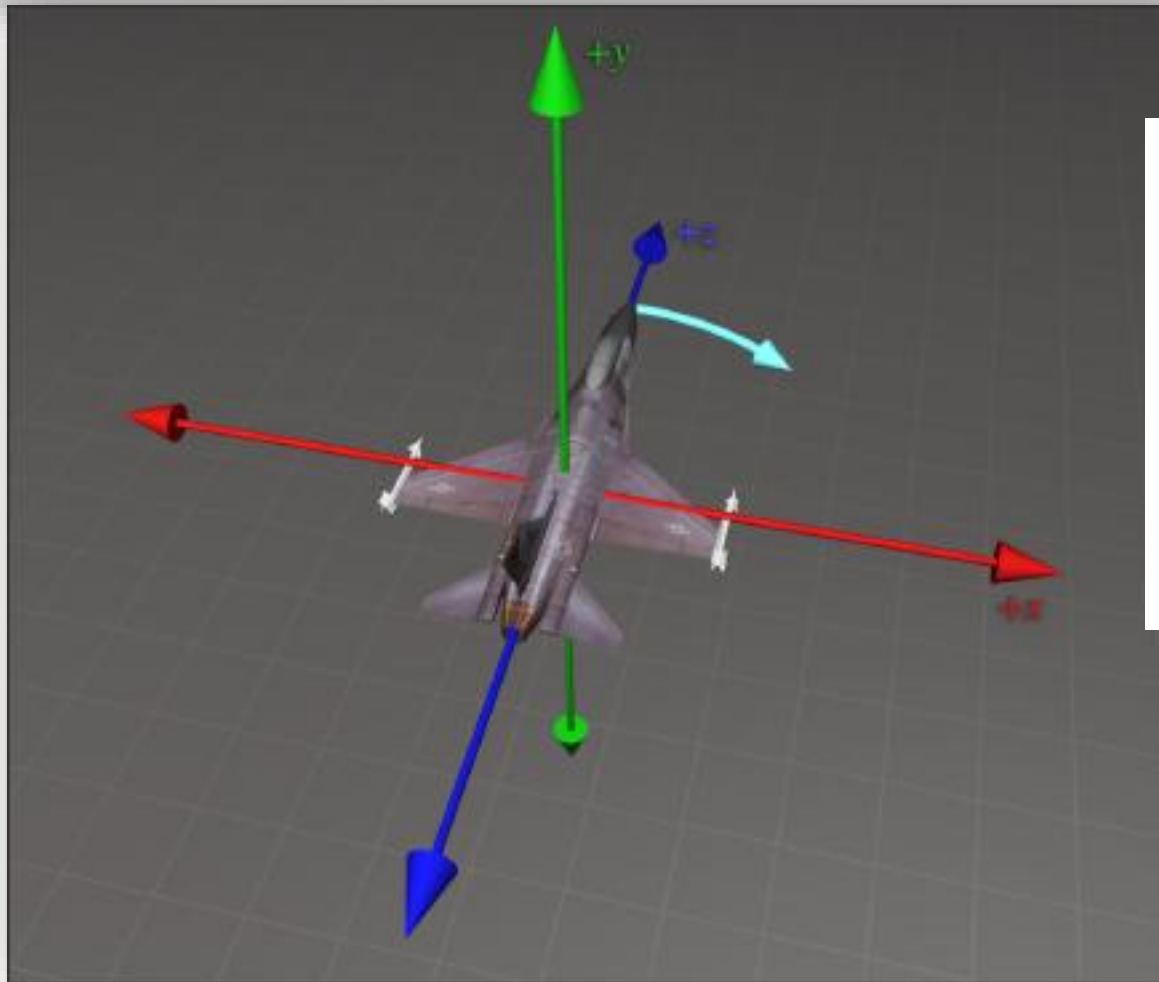


This is Important Because

Twisting an object changes its orientation.

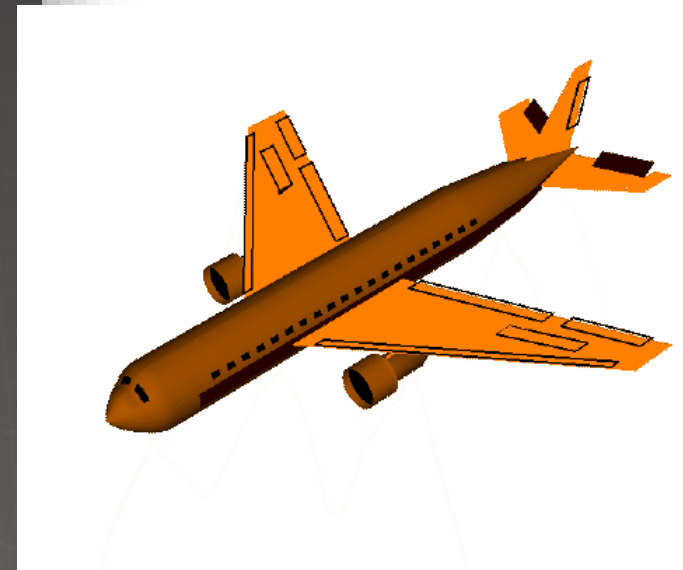
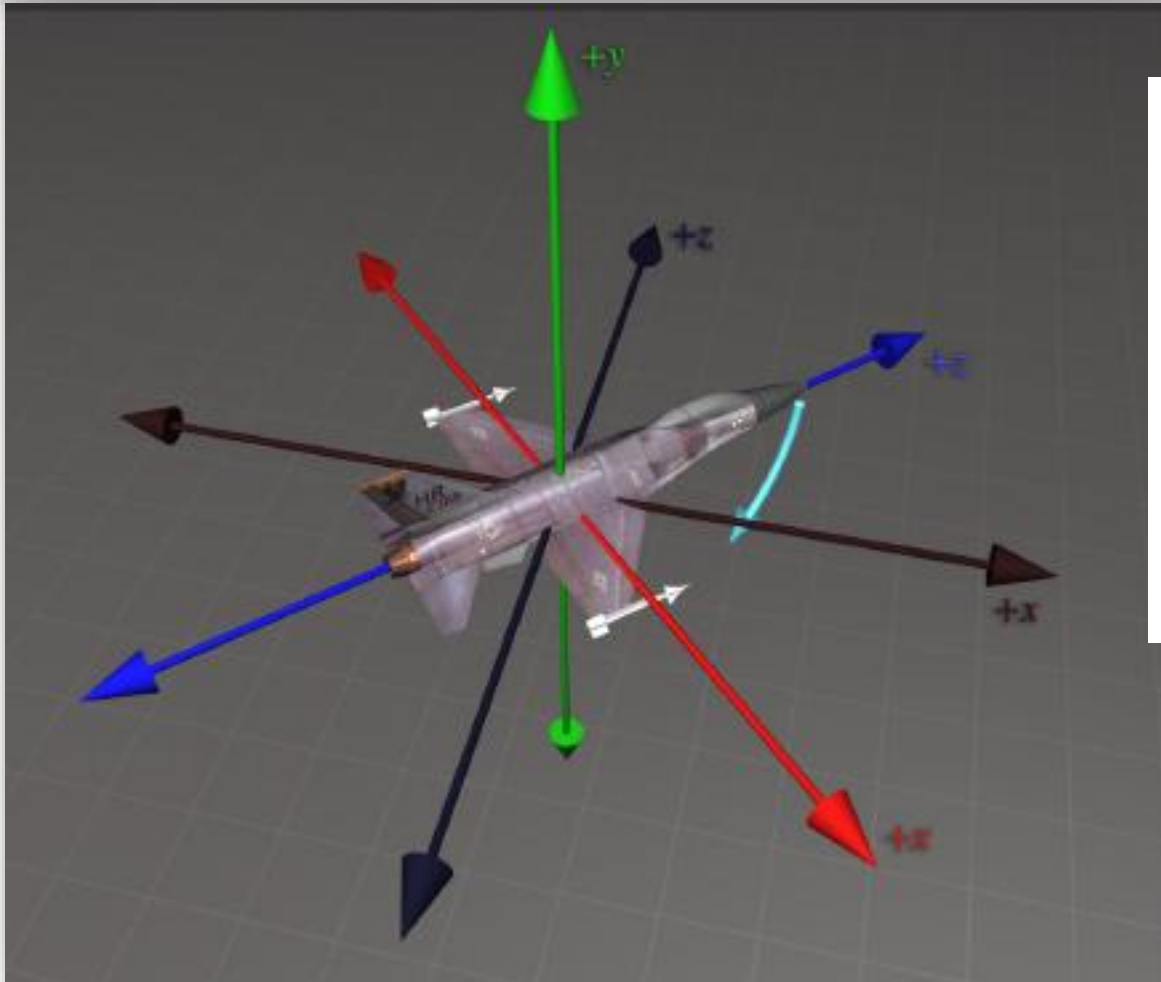


Heading – Eixo Vertical – Guinada



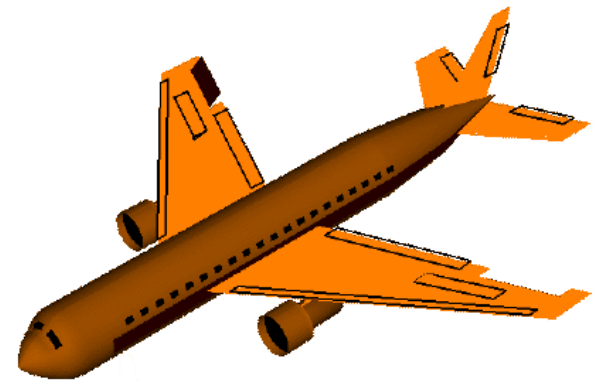
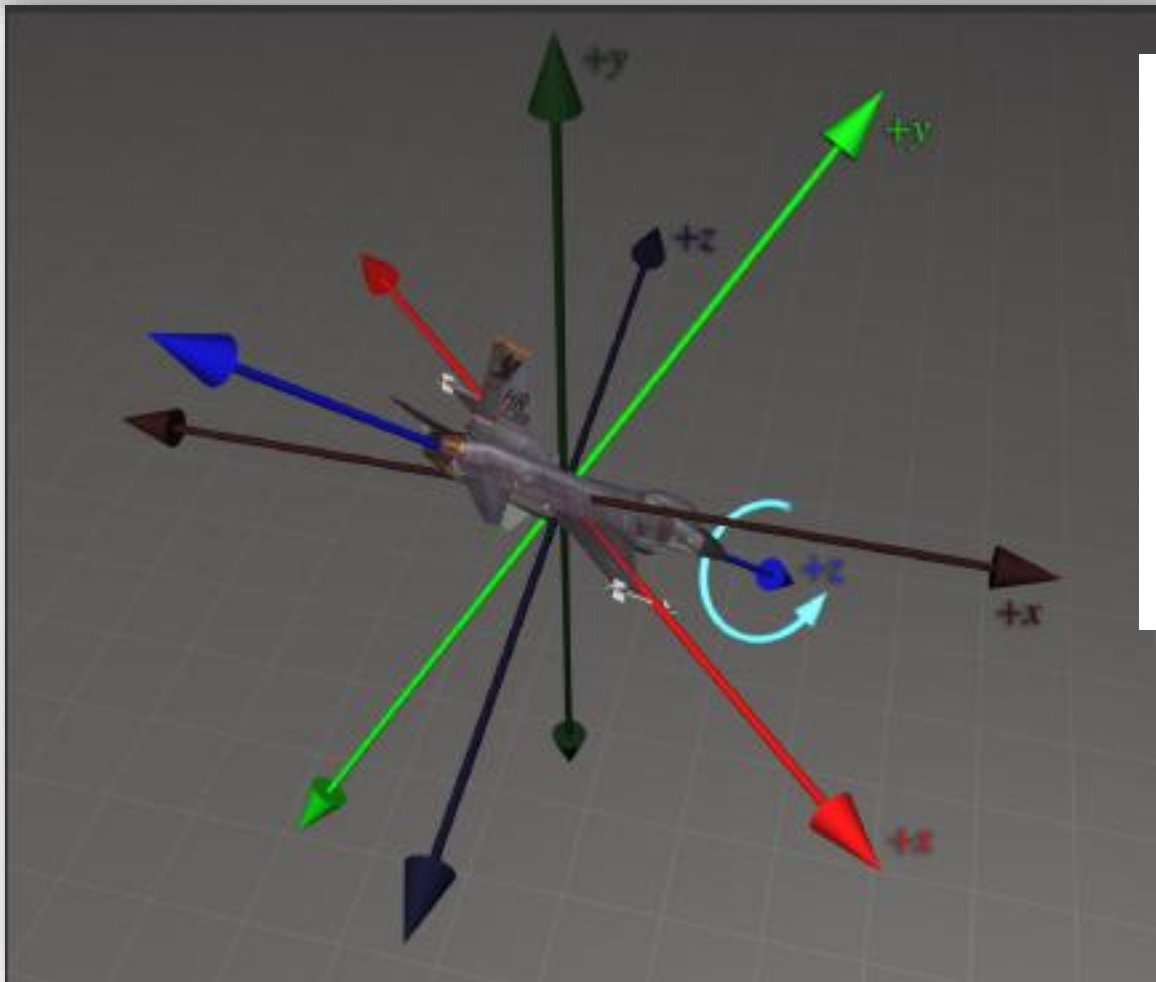
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